

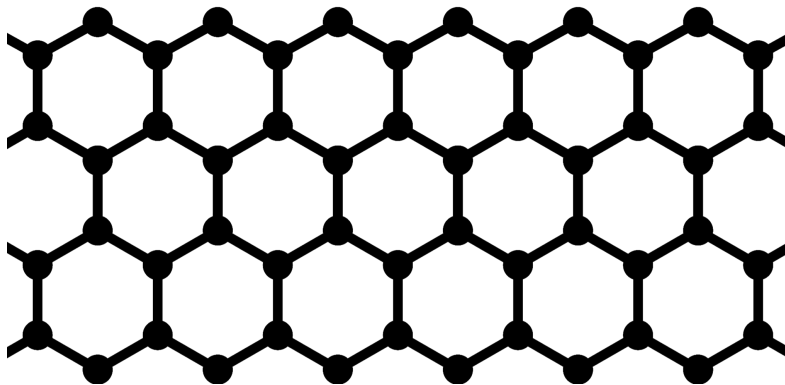
Predictive Maintenance: Machine Learning Anomaly Detection for Maritime Supply Chain Ship Operations

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 Project Repository



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1 Executive Summary

Ship engines record measurements such as speed, pressures, and temperatures. Unusual readings can signal wear, rising operational costs, or even impending failure. Spotting them early lets operators intervene before revenue loss, however, flagging too early squeezes profit margins with unnecessary investigations, so alerts should stay rare at around 1-5%. This challenge is shared across many industries, from identifying fraudulent transactions to faults in manufacturing processes.

This project compares common anomaly detection methods, asking the question of which approach identifies abnormal engine behaviour best across a multi-dimensional dataset of 20,000 records. Three methods were built and compared: a simple, baseline statistical rule which flags when values fall outside a normal range (Inter-Quartile Range), and two machine learning techniques, one that learns the optimal boundary between “normal” and “anomalous” records (One-Class Support Vector Machine), and the other which finds records that are unusually easy to segment from the rest (Isolation Forest). Each was calibrated to flag a few percent of readings with results visually compared on a collapsed 2D projection.

Results showed that ship engine anomalies are intertwined with normal values without an obvious pattern, with machine learning models being in good agreement with each other. Isolation Forest is recommended for its efficiency, performance, and relative explainability. The statistical baseline falls short on many counts due to equivalent treatment of all anomalies without consideration of correlation effects.

Two caveats apply: the 1-5% target is an assumption rather than empirically justified values, and readings were treated as independent snapshots rather than a progression through. A series of (near) anomalous records give a stronger indication than one highly abnormal record. As such, next steps would include adding temporal trends to flag when readings drift toward anomalous territory.

2 Introduction

Ship engines record data which can be leveraged to afford insight into anomalous operating conditions. Suboptimal conditions may accelerate wear, increased costs, and risk catastrophic failure and hence revenue loss.

Anomaly detection can highlight abnormal ship engine behaviour to enable early intervention and ease short term engine load. However, the rate of anomalies should be limited to 1–5% to ensure revenue is not capped by decreased efficiency (load limits/maintenance).

A dataset of 19535 records is provided [1]. Descriptive information generated by the Python `.describe()` function is seen in Table 1. Box plots for each feature are visualised in Figure 1. Pair-wise plots are seen in Figure 2. No duplicate or null-values found within the dataset for cleaning.

Statistic	engine_rpm	lub_oil_pressure	fuel_pressure	coolant_pressure	lub_oil_temp	coolant_temp
count	19535.00	19535.00	19535.00	19535.00	19535.00	19535.00
mean	791.24	3.30	6.66	2.34	77.64	78.43
std	267.61	1.02	2.76	1.04	3.11	6.21
min	61.00	0.00	0.00	0.00	71.32	61.67
50%	746.00	3.16	6.20	2.17	76.82	78.35
max	2239.00	7.27	21.14	7.48	89.58	195.53

Table 1: Summary Statistics of Engine Data

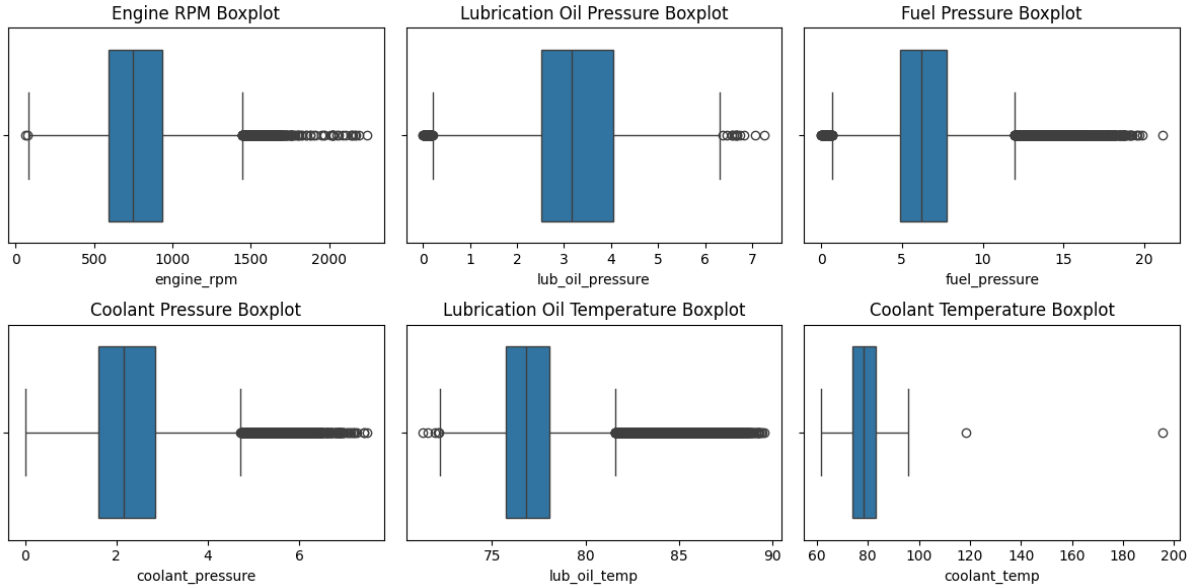


Figure 1: Boxplots for each feature in the original dataset. Significant anomalies detected beyond the whiskers of the plots, validating IQR method choice.

3 Methodology

A range of statistical and machine learning techniques are leveraged in this work. The overall workflow between methods remains similar, as visualised in Figure 3.

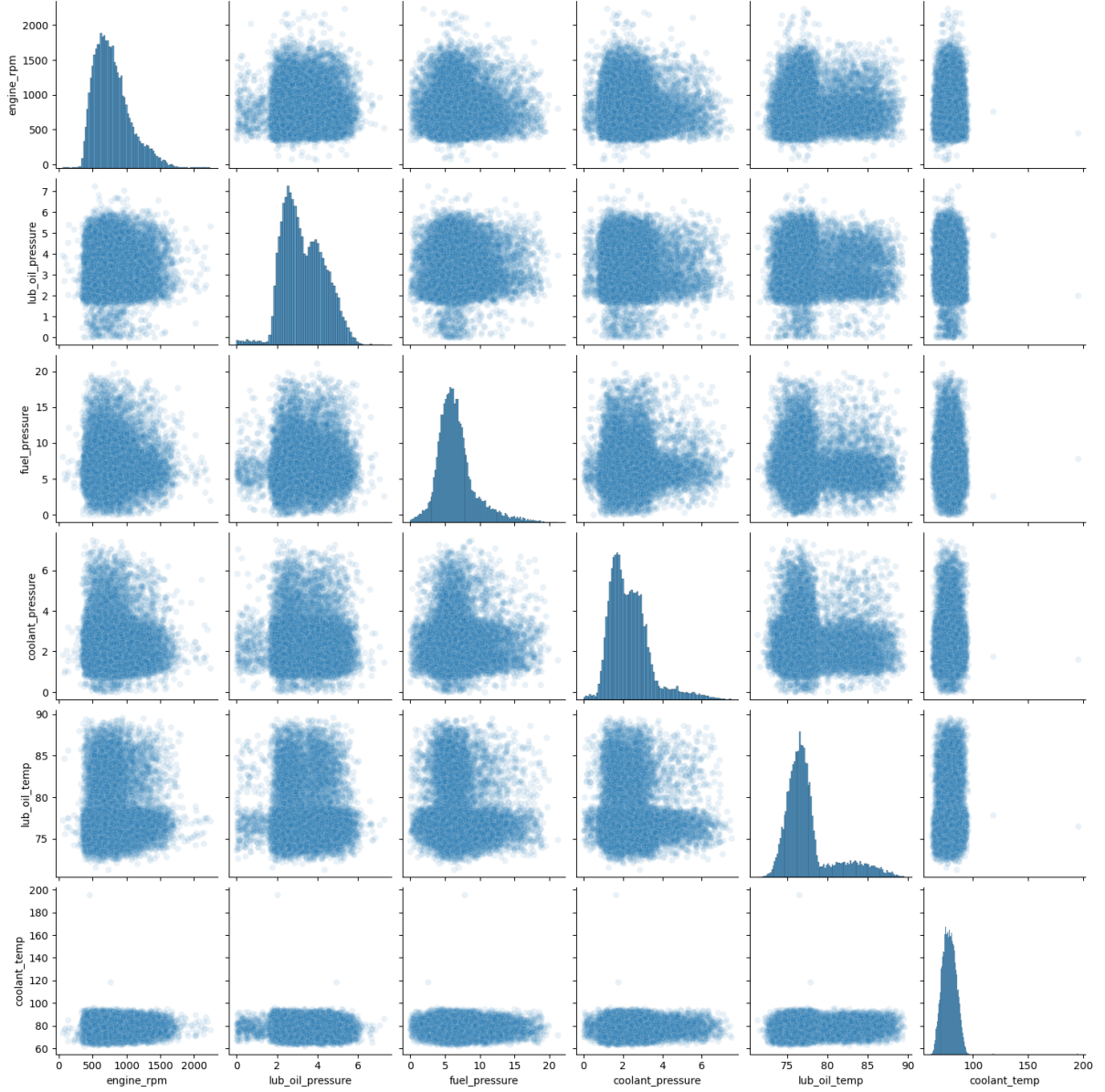


Figure 2: Feature pair-wise plots visualising relationships between features. Leading diagonal shows histogram distribution for each feature. No correlation visually seen between features in plots, verified by Spearman analysis (normality assumption void) with all absolute values less than 0.1.

3.1 Statistical Methods

3.1.1 Inter-Quartile Range (IQR)

The Inter-Quartile Range (IQR) method is statistically based: sorting each feature, then extracting Q_1 and Q_3 to find each's IQR. This is used to derive anomaly boundaries upon each feature: $1.5 \times IQR$ beyond Q_1 or Q_3 . As such, each record has up to six anomaly flags, summed to derive an anomaly sum score for each record. Anomaly scores range 0–3, with two anomalies selected as the minimum threshold as this corresponds to an anomaly rate of 2.16%, within the predefined 1–5% range.

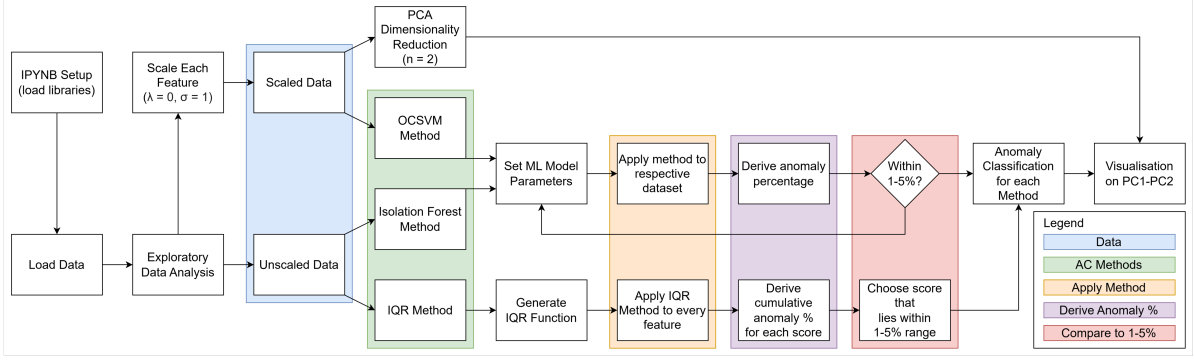


Figure 3: Overall workflow employed for anomaly detection. Three methods are employed: OCSVM, Isolation Forest, and IQR, each with anomaly percentage compared to the expected 1–5% range for model parameter validation, anomaly classification, and subsequent visualisation.

3.2 Machine Learning Methods

3.2.1 One-Class Support Vector Machine (OCSVM)

One-Class Support Vector Machine (OCSVM) is a kernel based anomaly detection technique which uses a kernel function to "warp" space such that linearly inseparable data can be separated by a hyperplane boundary. It is mainly controlled by two parameters: γ and ν , which control boundary tightness and approximate fraction of anomalies, respectively. This hyperplane boundary classifies points inside as normal, with everything outside as anomalous.

3.2.2 Isolation Forest

Isolation Forest is a decision tree based technique which assumes that anomalies are quickly isolated by random partitions across features, hence will appear closer to the root node of a tree than normal data points. Average isolation path length for each record is derived across the forest, with lower scoring records more likely to be anomalous. This method works to identify anomalies first, with everything else classified as "normal": opposite to OCSVM.

4 Results

PCA dimensionality reduction is performed on the scaled dataset for visualisation purposes so anomaly detection methods can be easily compared with as much information about the data's shape preserved. As such, data plots are shown on the PC1-PC2 projection plane. PC1 and PC2 have respective explained variances of 19.0% and 17.7%, leaving over 60% of variance unexplained on these axes. This acts as a limitation of PCA visualisation for this dataset as significant information is lost when plotted in 2D. Other techniques, like t-SNE, may reveal more useful structure but lie outside of scope.

Multiple models of Isolation Forest and OCSVM were generated with different machine learning (ML) parameters, however, Figure 4 shows only parameter combinations which align with the assumed anomaly rate of 1–5%, with respective anomaly percentages seen in Table 2.

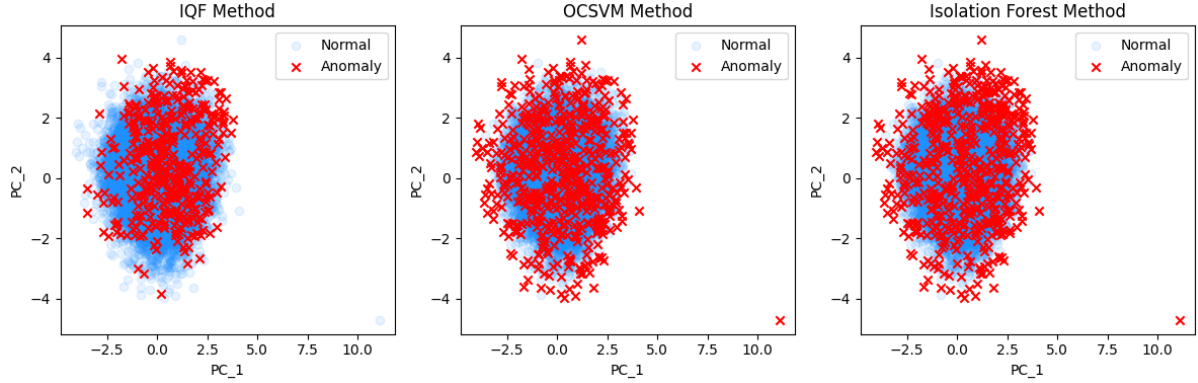


Figure 4: Comparison of performance of anomaly detection methods: IQF, OCSVM, and Isolation Forest. Anomalies, red crosses, are overlaid atop translucent blue circles, normal data points. All methods afford a similar overall trend that anomalies are intertwined with normal values when projected upon PC1-PC2 with no obvious discernable pattern. OCSVM parameters: $\gamma = 0.3$, $\nu = 0.03$; Isolation Forest parameter: $N_t = 100$, $\nu = 0.03$. [Note: ν used for Isolation Forest contamination since similar concept.]

Method	Anomaly Percentage / %
IQR	2.16
OCSVM	3.02
Isolation Forest	3.00

Table 2: Anomaly Percentages for Methods

All methods afford similar anomaly distribution, however, ML methods are more similar to each other than they are to the statistical IQR method.

There is a data point in the bottom right of each plot in Figure 4 which the ML methods classify as anomalous, whereas IQF classifies as normal. At first glance, this point seems anomalous as it sits far from the main "egg-shaped" cluster. However, IQR acts feature-wise and the dataset exists in a six-dimensional feature-space. Therefore, this data point may retain non-anomalous character from other features which span directions orthogonal to PC1 and PC2 (the other approximately 60% of unexplained variance). Otherwise, the IQF method shows fewer anomalies around the periphery of the "egg-shaped" cluster compared to the ML methods, with a higher anomaly concentration in the core of the cluster.

ML methods are visually similar with difficulty arising trying to discern differences. Core cluster anomaly spread is the main differentiator where Isolation Forest shows more anomaly "gaps", where the blue of normal data points shows through, yet alone this is not enough to prefer one method to another.

5 Conclusion

Each method: IQR, OCSVM, and Isolation Forest, provide similar anomaly classification information. After PCA dimensionality reduction, visualisation of each method shows that anomalies are closely intertwined with normal data across the PC1-PC2 projection plane with no obvious pattern identified.

IQF provides a more explainable method so may be preferred for stakeholders without knowledge or trust in ML methods. However, IQF fails to identify the importance and weighting of anomaly type, treating each anomaly identically. That is, a "slight" anomaly is treated the same as an "extreme" anomaly, and anomaly interaction is not considered (i.e. the combination of an anomaly in feature x and y might be more contextually relevant than the combination of anomalies across features a , b , and c). This could be addressed with an additional aggregation/weighting model such that overall IQR anomaly scoring is not a simple summation of anomaly occurrence, yet this could require enhanced domain knowledge and stakeholder engagement.

With respect to ML methods, OCSVM performs similarly to Isolation Forest but the latter acts to identify anomalies first, making it a more efficient method than OCSVM. Moreover, Isolation Forest retains moderate explainability with basic knowledge of decision trees and forests. As such, Isolation Forest is both more explainable and efficient so is chosen as the preferred anomaly detection method for this dataset given the aforementioned shortcomings of IQR.

However, ship data is temporal in nature, i.e. sequential measurements are likely highly correlated assuming acquisition frequency is high enough. This could give more useful information via derivatives to understand the rate of change of variables as they tend towards an anomaly boundary. Moreover, the anomaly detection limit is not empirically justified, rather, it acts as a subjective value that model parameters are tuned for.

Overall, an Isolation Forest model with $N_t = 100$ and $\nu = 0.03$ could be used to classify future ship engine operations as anomalous or not. This gives useful insight into whether a catastrophic failure may occur and should be combined with temporal reasoning and monitoring to flag when anomaly rates rise too quickly.

References

- [1] D Mohakul. Predictive maintenance on ship's main engine using ai. *IEEE Dataport*, 2022. URL <https://dx.doi.org/10.21227/g3za-v415>. Accessed 29/03/2026.